

Fungal-bacteria interactions provide shelter for bacteria in Caesarean section scar diverticulum

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Abstract

Caesarean section scar diverticulum (CSD) is a significant cause of infertility among women who have previously had a Caesarean section, primarily due to persistent inflammatory exudation associated with this condition. Even though abnormal bacterial composition is identified as a critical factor leading to this chronic inflammation, clinical data suggests that a long-term cure is often unattainable with antibiotic treatment alone. In our study, we employed metagenomic analysis and mass spectrometry techniques to investigate the fungal composition in CSD and its interaction with bacteria. We discovered that local fungal abnormalities in CSD can disrupt the stability of the bacterial population and the entire microbial community by altering bacterial abundance via specific metabolites. For instance, *Lachnellula suecica* reduces the abundance of several *Lactobacillus spp.*, such as *Lactobacillus jensenii*, by diminishing the production of metabolites like *Goyaglycoside A* and *Janthitrem E*. Concurrently, *Clavispora lusitaniae* and *Ophiocordyceps australis* can synergistically impact the abundance of *Lactobacillus spp.* by modulating metabolite abundance. Our findings underscore that abnormal fungal composition and activity are key drivers of local bacterial dysbiosis in CSD.

eLife assessment

This study has uncovered some interesting findings about the fungal composition and its interaction with bacteria in Caesarean section scar diverticulum (CSD). While the study's findings are **valuable** and with translation possibilities, the strength of the conclusions obtained is **incomplete** due to the small sample size and methodological issues indicated by the reviewers such as the lack of controls and the location of samples analyzed.

Introduction

Caesarean section (CS) is a prevalent surgery worldwide, and its rate has increased in recent decades [1]. Although CS can significantly reduce dystocia and stillbirths [2], Caesarean section scar diverticulum (CSD) affect about 19.4-88% of women receiving this operation [3]. CSD emerges from poor healing of the local uterine incision, forming a depression or cavity that connects with the uterine cavity [4]. Recently, CSD has drawn widespread attention because of its potential damage to subsequent fertility. For example, Gurol and colleagues found that the possibility of subsequent pregnancy decreases by an average of 10% after CS relative to a previous vaginal delivery [5]. A niche can reduce the chances of embryo implantation and increase the likelihood of spontaneous miscarriages if the implantation occurs close to or in the CSD [6]. Our previous studies have shown that the persistent effusion of CSD is a key cause of failed embryo implantation [7].

The unique microbial community composition in the female reproductive tract is essential in maintaining female reproductive health [8-10]. Our previous studies have demonstrated that the abnormal alterations in the local microbiota of Cesarean scar defects (CSD) cause continuous leakage through local inflammation and immune imbalance [11]. Further investigations of ours revealed the underlying mechanism demonstrating that abnormal bacteria in CSD deplete protective fatty acids and generate N-(3-hydroxy-eicosanoyl)-homoserine lactone, thus leading to promoting apoptosis of vascular endothelial cells and endometrial epithelial cells, ultimately impairing women's reproductive capacity [12]. Nonetheless, the causative factors behind bacterial dysbiosis in complex microbial communities remain poorly understood.

Fungi and bacteria, as integral components of the human microbiome, establish complex interactions with each other and the host. The collection of genomes and genes carried by fungal species coexisting within a specific environmental or biological niche is commonly called as the "mycobiome" [13]. However, in CSD, the mycobiome remains poorly understood, particularly in comparison to the microbiome. The interaction between microbial communities in the reproductive tract, particularly bacteria and fungi, is critical in maintaining female reproductive health [14]. Therefore, this study aims to examine the composition and function of fungi in CSD, as well as their interactions with bacteria, to provide a more comprehensive understanding of the role of microbial communities in CSD. This study also aims to identify potential therapeutic approaches to enhance fertility by building on current research.

Methods

Participant recruitment and sample collection

The subjects were recruited at the Reproductive Medical Center of the Sixth Affiliated Hospital of Sun Yat-Sen University. The inclusion and exclusion criteria were previously described in a previous study [12]. All participants were fully informed, volunteered to participate in this study, and signed informed consent forms. All study processes were reviewed and approved by the Sixth Affiliated Hospital Ethics Committee of Sun Yat-Sen University (IRB no. 2019ZSLYEC-005S).

The sample collection procedure is as described above [12]. In short, after cleaning the external genitalia and vagina, a sterile disposable swab is inserted into the cervical canal, and the swab is rotated five times to collect the sample fully. The sample is rapidly transferred to liquid nitrogen for quenching and then stored at -80°C.

DNA extraction and metagenomic sequencing

According to the manufacturer's instructions, the total DNA was isolated from the sample using QIAamp® Fast DNA Mini Kit (Qiagen, Hilden, Germany). The DNA concentration and integrity were evaluated by NanoDrop2000 spectrophotometer (Thermo Fisher Scientific, Waltham, MA, USA) and agarose gel electrophoresis. The DNA was fragmented with S220 Focused-ultrasonicators (Covaris, USA) and purified with Agencourt AMPure XP magnetic beads (Beckman Coulter Co., USA). Then, the library was constructed using the TruSeq Nano DNA LT Sample Preparation Kit (Illumina, San Diego, CA, USA) according to the manufacturer's instructions. The sequencing was performed using Illumina NovaSeq 6000.

Raw Data Preprocessing

Parallel computing in research is implemented with GNU Parallel [15].

The sequencing raw data was stored in FASTQ files. After trimming and filtering with Trimmomatic (v0.36) [16], the clean paired-end reads were aligned to the host genome (hg38) with bowtie2 (Presets: --very-sensitive-local) [17], and the matched reads were discarded. The clean reads were assembled into contigs using MEGAHIT software (--k-list 21,33,55,77) [18] with a filtering threshold of 500 bp. The ORFs were predicted using Prodigal software [19] and were translated into amino acid sequences. CD-HIT software [20] was used to remove redundancy from the ORF predictions of each sample and mixed assembly with default parameters, and to obtain non-redundant initial Unigenes. The clustering was performed with default parameters of 95% identity and 90% coverage, and the longest sequence was selected as the representative sequence. Bowtie2 software was used to align the clean reads of each sample to the non-redundant gene set (95% identity), and the abundance information of genes in each corresponding sample was calculated from the number of reads and the length of the genes.

Taxa and functional annotation

The representative sequences of the redundant gene set were aligned with NCBI's NR database using DIAMOND software [21], and annotations with $e < 1e-5$ were selected to obtain proteins with the highest sequence similarity, thereby obtaining functional annotations and species annotations. Species present in only <10% of samples were excluded. The Valencia software [22] was used to perform community state type (CST) assignment for each sample. The abundance of species is calculated using the sum of the gene abundance corresponding to the species. LEfSe (Linear discriminant analysis Effect Size) [23] was used to screen differentially abundant species by using MicrobiotaProcess R package [24]. The first comparison uses the Kluskal-Wallis test, the second comparison uses the Wilcoxon test, and finally LDA is used to determine the difference, and the p value is corrected using *fdr*. The calculation and visualization of species diversity and abundance are executed by the EasyMicroPlot R package [25] and EasyAmplicon pipeline [26]. Annotations were performed using the KO, COG, and CAZy databases to explore changes in enzyme family related to the microbiome. Differential functional analysis was conducted using the STAMP software [27].

Construction of co-occurrence network

In network construction, we used more stringent species filtering criteria and only included species detected in 50% of samples in the analysis. The construction and random robustness comparison of fungal and bacterial-fungal co-occurrence networks were executed by the ggClusterNet R package [28]. Spearman correlation analysis was used to calculate correlations, the cutoff value of the correlation coefficient was set to 0.6, and p values were corrected using FDR. The stability analysis of the network is implemented by the Robustness.Random.removal function of the ggClusterNet R package with the default parameters.

Metabolite detection using liquid chromatography (LC) and mass spectrometry (MS)

A separation buffer (methanol/acetonitrile/water [2:2:1]) was used to extract metabolites from cervical swabs. Ten μ l was taken from each sample and mixed to form a quality control (QC) sample [29]. This was used to evaluate stability during the experiment. The metabolite signals in each sample were detected using liquid chromatography (LC) and mass spectrometry (MS) (ACQUITY UPLC I-Class plus, Waters). Progenesis QI v2.3 software was used for qualitative analysis of metabolites, with parameters of 5 ppm precursor tolerance, 10 ppm product tolerance, and 5% product ion threshold (Nonlinear Dynamics, Newcastle, UK). All output data were normalized using internal standards, and the results were displayed as peak values (test sample peak area/internal standard sample peak area). Compound identification was performed using human metabolome database (HMDB), Lipidmaps (V2.3), Metlin, EMDB, PMDB, and a self-built database based on mass-to-charge ratio (M/z), secondary fragmentation, and isotope distribution.

The screening of differential metabolites was performed using the OPLS-DA (orthogonal partial least squares discriminant analysis) method, with a VIP value >1 for the first principal component and a P value <0.05 for T-test as the threshold. MetPA was used for differential metabolic pathway analysis [30]. We used Spearman correlation analysis to evaluate the correlation between the microbial community and metabolites. The network was visualized using Cytoscape software [31].

Results

Recruitment of participants and metagenomic sequencing

Forty-eight participants were included in this study, including 24 in the CSD group and 24 in the CON group. The clinical characteristics of the participants are shown in **Table 1**. To better characterize the composition of the cervical microbiota, we performed ultra-high-depth metagenomic sequencing. The effective data volume of each sample ranged from 14.25 to 18.63 G, the N50 statistical distribution of contigs ranged from 230 to 245 bp, and the number of ORFs in the gene catalogue (non-redundant gene set) constructed after redundancy removal was 281,107.

Composition and characteristics of bacterial communities

Alpha diversity based on Pielou index (**Fig. 1 A**) and Shannon index (**Fig. 1 B**) calculations showed that species richness and evenness in the CSD group were significantly higher than those in the control group (T-test). Beta diversity based on the Bray-Curtis distance (**Fig. 1 C**) indicated that the distances between samples within the CSD group were greater than those in the CON group (PERMANOVA test).

The composition of CST community types in the CSD group and CON group differed. The proportion of CST III decreased in the CSD group, while the more unstable CST IV-B and CST IV-C increased (**Fig. 1 D**). The CST community types in the CON group were dominated by *Lactobacillus* spp. (**Fig. 1 D**). CST subtype analysis revealed more detailed community types. In CST I (dominated by *L. crispatus*) and CST III (dominated by *L. iners*), the CSD group was mainly distributed in the B subtypes with a lower abundance of *L. crispatus* and *L. iners* (**Fig. 1 E**). The stacked bar chart of species composition shows the species composition characteristics of each sample (**Fig. 1 F**). One hundred and sixty-two differential species were identified between the two groups. The differential species analysis indicated that *Bifidobacteriaceae* bacterium spp. and *Gardnerella* spp. were significantly higher in the CSD group than in the CON group (**Fig. 1 G**).

	CON Group (n=24)	CSD Group (n=24)	<i>p</i> value
AMH (ng/ml) (mean (SD))	3.69 (2.09)	3.00 (2.32)	0.288
BMI (mean (SD))	21.04 (2.25)	22.25 (2.70)	0.1
Basal FSH (IU/L) (mean (SD))	6.99 (1.42)	7.70 (1.88)	0.144
Basal LH (IU/L) (mean (SD))	5.51 (2.97)	5.91 (3.55)	0.676
Basal E2 (pg/ml) (mean (SD))	47.42 (68.52)	44.49 (29.46)	0.848
infertile years (mean (SD))	3.54 (2.19)	4.54 (2.90)	0.184

Table 1.

Clinical characteristics of the participants included in the study.

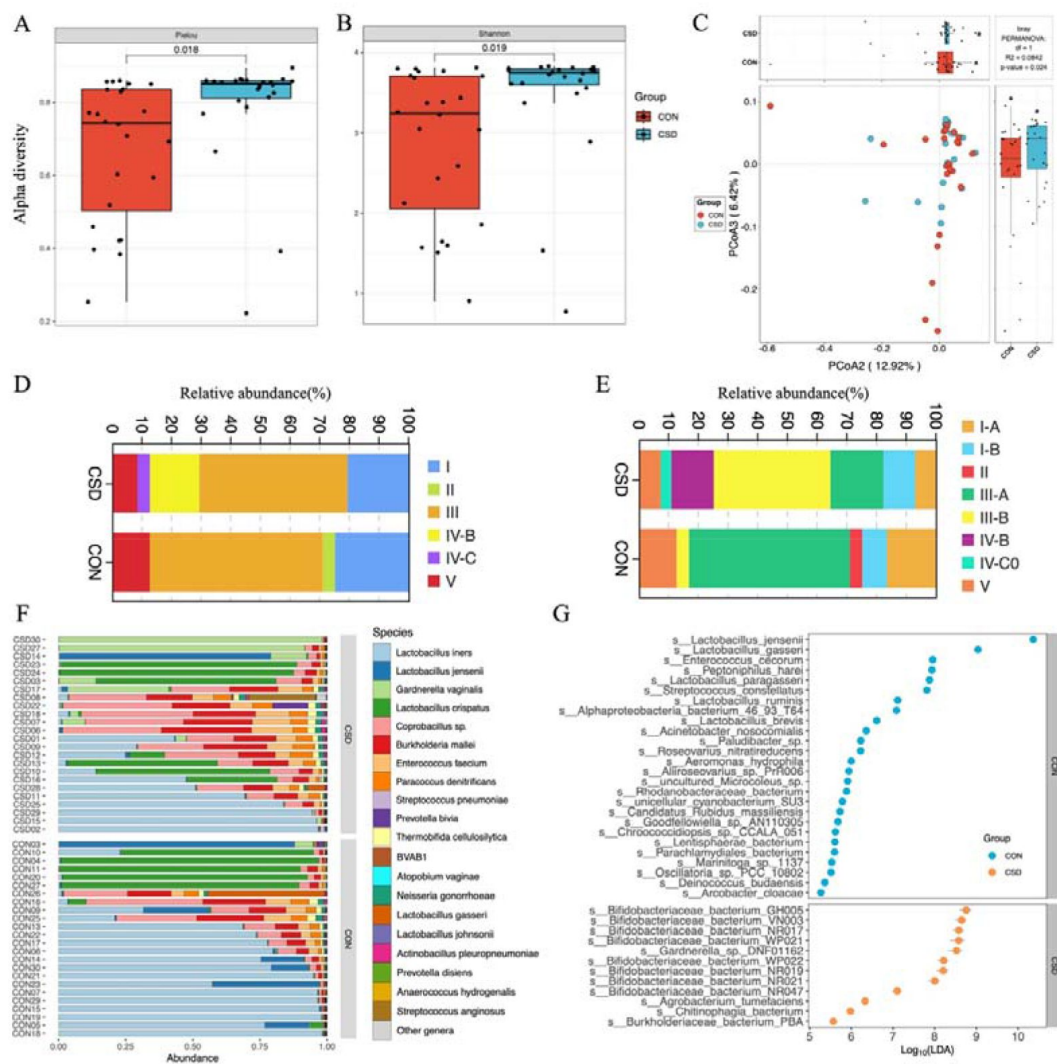


Fig 1.

Bacterial community structure and composition characteristics. Alpha diversity based on Pielou index (A) and Shannon index (B); (C) Beta diversity based on the Bray-Curtis distance; Stacked graph of CST (D) and subCST (E) compositions; (F) Stacked plot of species composition; (G) Scatter plot of differentially abundant species between CSD and CON groups.

Composition and characteristics of fungal communities

A total of 431 fungal species were annotated (Supplementary table 1). There was no difference in α diversity based on the Pielou index (Fig.2 A) and Shannon index (Fig.2 B) between the two fungal communities. β diversity suggested similar distances between samples within each group (Fig.2 C). Regarding species composition, the composition of the two groups was similar at the phylum (Fig.2 D) and species (Fig.2 E) levels. The tree map revealed the relationships among fungal phylum (Fig.2 F). Forty-two differential species were identified between the two groups, and the top 10 species are shown in Fig.2 G.

The co-occurrence network among microbial community

We constructed co-occurrence networks for fungal species in the CSD and CON groups (Fig.3 A). The network in the CSD group had fewer connections and weaker random robustness than the CON group (Fig.3 B), indicating greater fragility of the CSD fungal co-occurrence network. Subsequently, a cross-domain network between bacteria and fungi was constructed (Fig.3 C). The connection between bacteria and fungi in the CSD group was reduced compared with that in the CON group. At the same time, the CSD group had weaker random robustness than the CON group (Fig.3 D). We selected species with a sample coverage rate greater than or equal to 60% for Spearman correlation analysis and visualized the data using heatmaps (Fig.3 E) and networks (Fig.3 F). The results showed that *Lachnellula suecica*, *Arthrobotrys oligospora*, and *Piptocephalis cylindrospora* have close relationships with bacteria (Fig.3 F). *Piptocephalis cylindrospora* has a close symbiotic relationship with *Lactobacillus jensenii* (Fig.3 F).

Functional gene analysis

Functional gene analysis indicates significant differences in the functional gene composition of microbial communities between the CSD and CON groups (Fig.S1 A). The CSD group has two significantly higher KEGG modules than the CON group, including M00142 (NADH: ubiquinone oxidoreductase, mitochondrion) and M00151 (cytochrome bc1 complex respiratory unit) (Fig.S1 B). The CSD group has significantly increased activity in the inflammatory mediator regulation of TRP channels and platelet activation, while the activity of the galactose metabolic process has decreased (Fig.S1 C). The CAZy annotation results indicate that the activity of GH73 (glycoside hydrolase family 73) in the CSD group has decreased (Fig.S1 D).

Untargeted metabolomics revealed unique metabolic characteristics

LC/MS untargeted metabolomics was performed on 40 of the subjects. The heatmap shows differences in the distribution of metabolites between the two groups (Fig.4 A). Enrichment analysis of signaling pathways suggests that the CSD group has significantly increased activity in the Thermogenesis and Pentose phosphate signaling pathways, while the activity in Steroid biosynthesis is significantly decreased (Fig.4 B). Spearman correlation analysis suggests a close correlation between changes in the metabolite contents of *Lachnellula suecica* and *Arthrobotrys oligospora* (Fig.4 C). At the same time, we also explored the relationship between bacterial and metabolite content changes (Fig.S2).

Integrating bacterial, fungal, and metabolite data, we found that the fungus *Lachnellula suecica* has a mutual regulation relationship with *Lactobacillus spp.* through GoyaglycosideA and Janthitrem E (Fig. 5). At the same time, *Lachnellula suecica* occupies a central position in the entire regulatory network and is closely related to changes in metabolites and bacterial composition. *Clavispora lusitaniae* and *Ophiocordyceps australis* are also fungal species that are closely related to changes in the abundance of *Lactobacillus spp.* (Fig. 5).

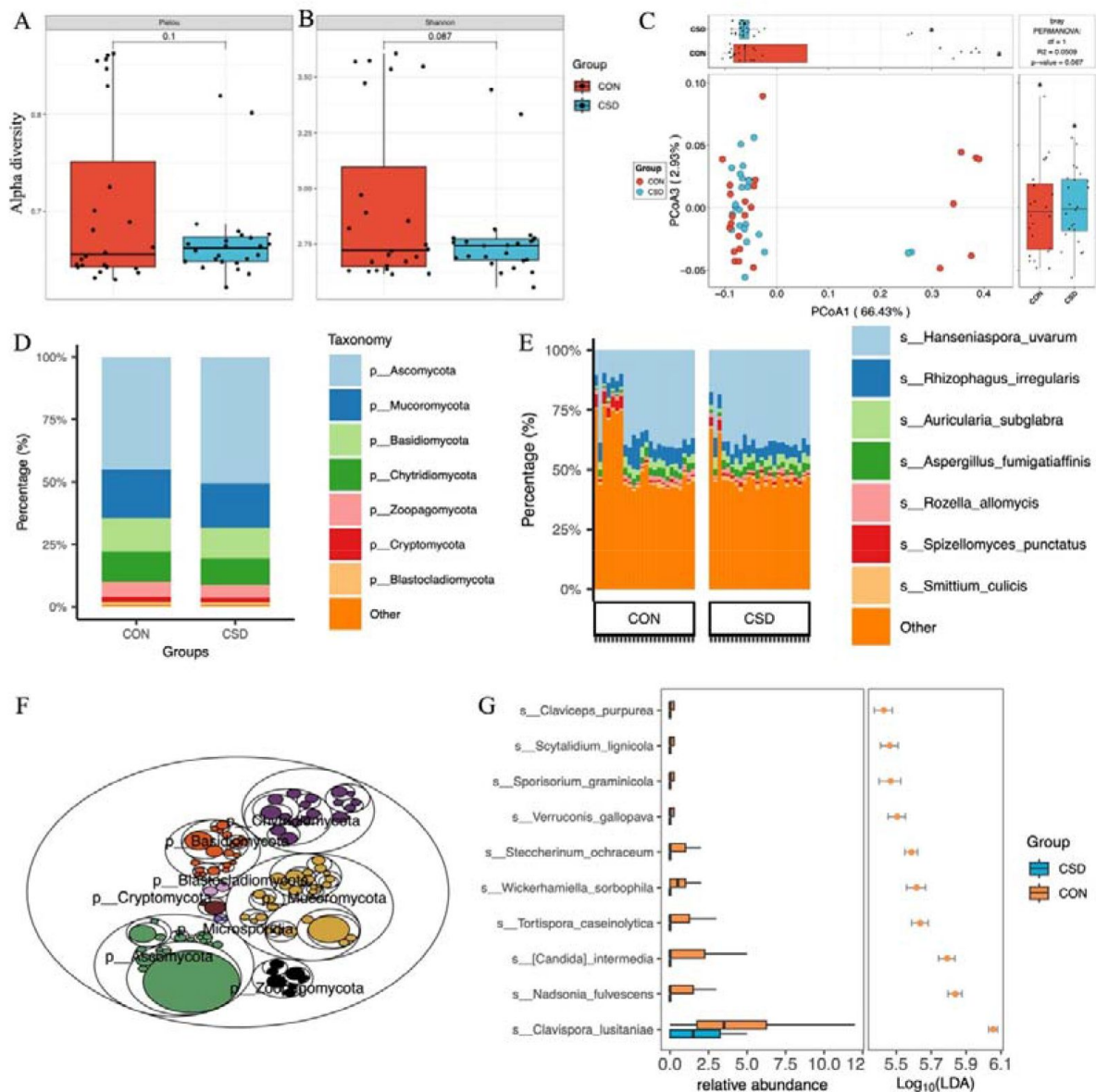


Fig 2.

Fungal community structure and composition characteristics. Alpha diversity based on Pielou index (A) and Shannon index (B); (C) Beta diversity based on the Bray-Curtis distance; Stacked graph of phylum level (D) and species level (E) compositions; (F) The tree map of fungal community; (G) Box plot of differentially abundant species between CSD and CON groups.

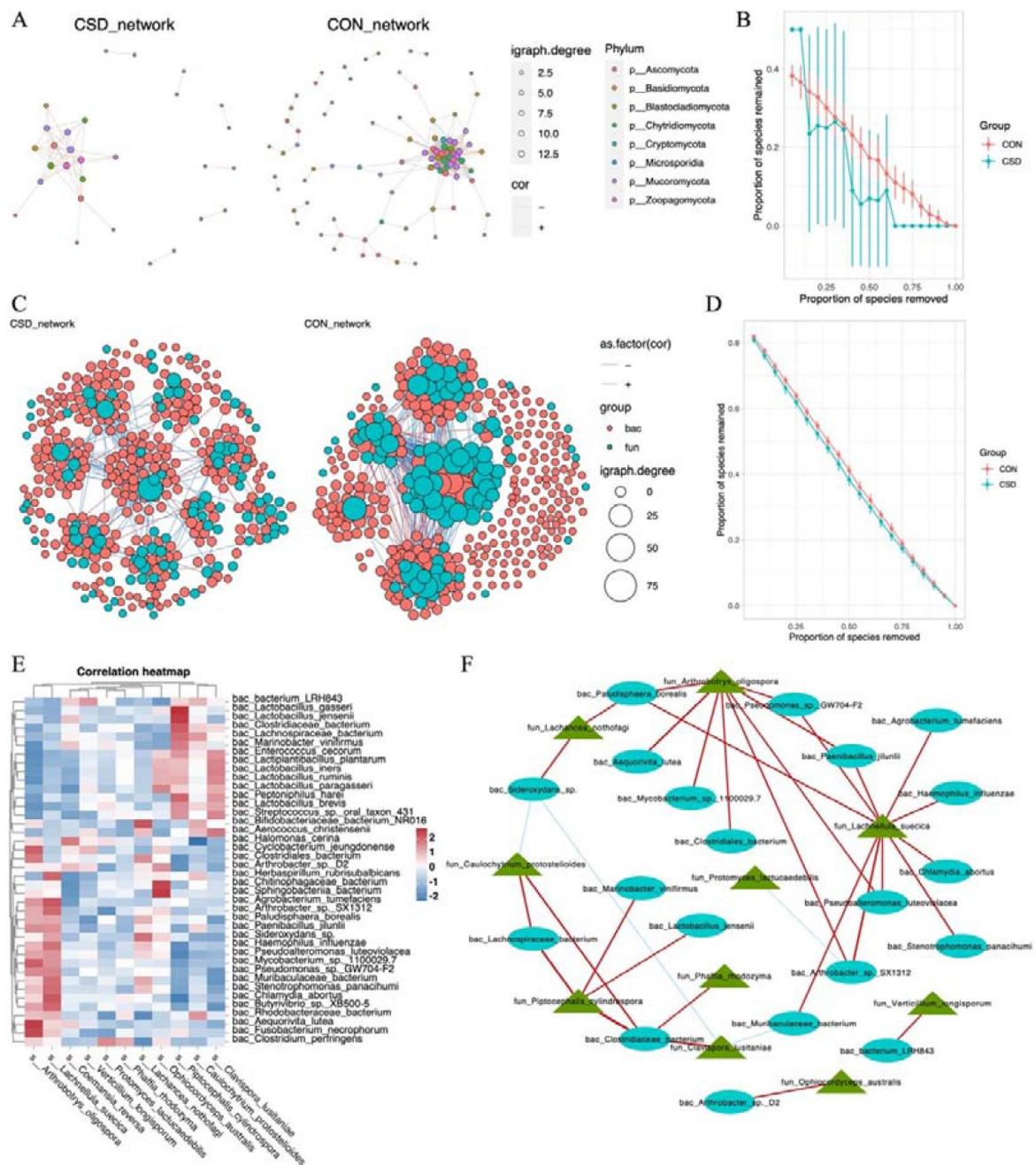


Fig 3.

Interaction between bacteria and fungi. (A) Co-occurrence network of fungal communities; (B) Comparison of the random robustness of co-occurrence network of fungal communities between CSD and CON groups; (C) Interaction network between bacteria and fungi; (D) Comparison of the random robustness of bacterial-fungal interaction network between CSD and CON groups; (E) Heatmap of the correlation between bacteria and fungi; (F) Network of correlations between bacteria and fungi ($R \geq 0.6$, $p \leq 0.05$).

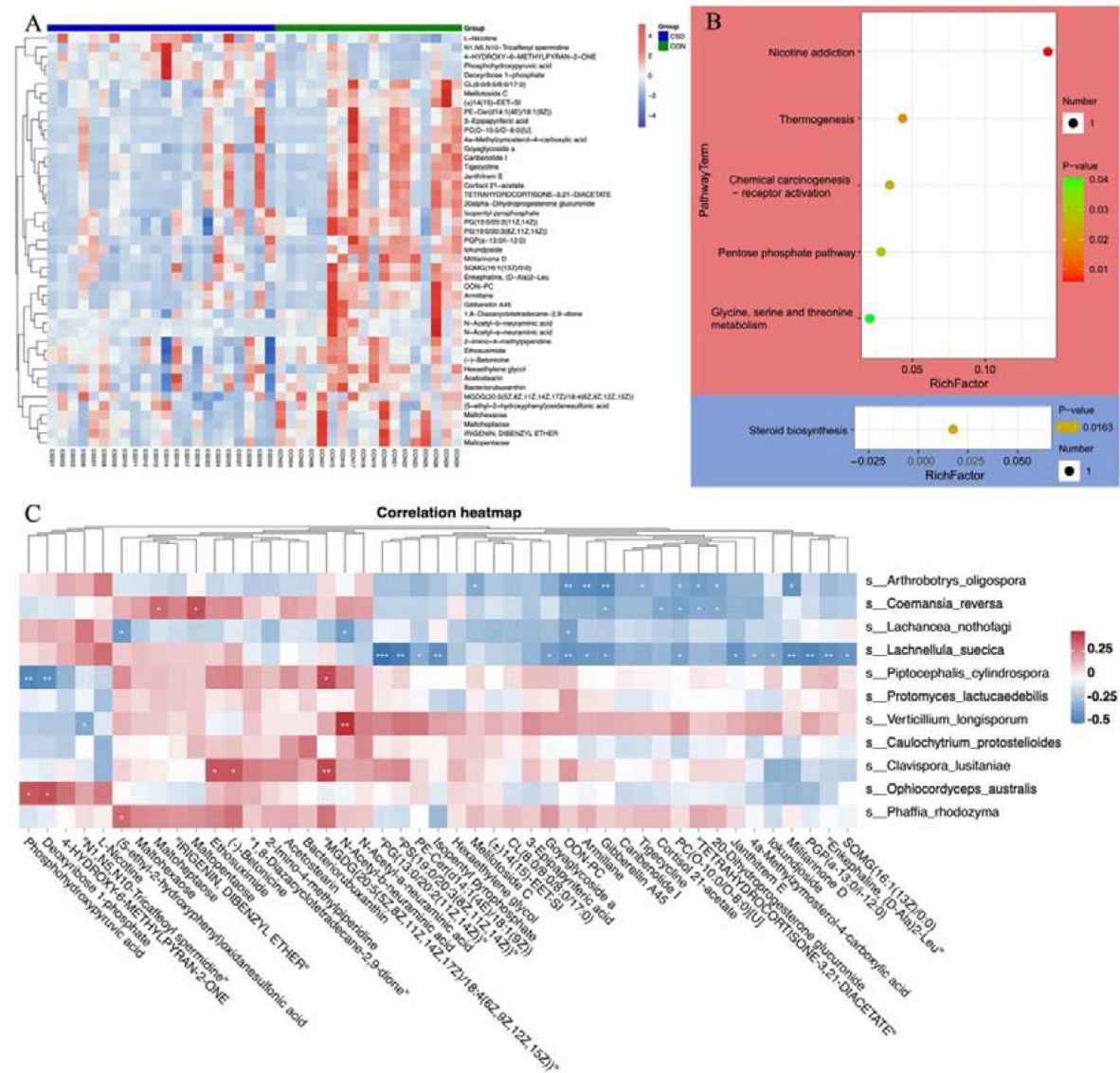


Fig 4.

Metabolite composition and functional characteristics. (A) Heatmap of differentially abundant metabolites between CSD and CON groups; (B) Dot plot of KEGG enrichment analysis of metabolites. Red represents upregulation, blue represents downregulation; (C) Heatmap of the correlation between fungi and metabolites.



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Discussion

Caesarean section scar diverticulum (CSD) significantly affects female fertility, posing a challenge to women who wish to conceive. Our research team conducted a preliminary investigation on bacteria-host interactions [12]. Current findings suggest that fungi play crucial roles in maintaining bacterial community and microbiome stability [32, 33]. This study explores the intricate fungal-bacterial interactions within the microbiome of CSD, building on our team's previous research findings.

We achieved more precise bacterial community compositions by implementing higher-resolution metagenomics than our previous study [12]. *Lactobacillus jensenii* of the *Lactobacillus* genus exhibited the greatest discrepancy between the two groups. This species has been linked to integral aspects of women's reproductive health. It has been demonstrated that a reduction in *L. jensenii* abundance has a close correlation with early-stage embryo arrest [34]. Furthermore, research suggests that *L. jensenii* may hold potential in facilitating vertical transmission from mother to infant [35].

Fungi and bacteria, as integral components of the human microbiome, establish complex interactions with each other and the host. The characteristics of fungal communities in CSD and CON group are similar, but there are differences in species composition. *Clavispora lusitaniae* is the species with the largest difference between the two groups. In addition, the co-occurrence network of fungal communities and the stability of the bacterial-fungal cross-domain network in the CSD group are poorer than those in the CON group, indicating that the cervical microbiota stability is disrupted in CSD, resulting in local microbial-immune imbalance. A previous study confirmed that persistent exudation in CSD is a crucial factor affecting embryo implantation [7]. Several studies have shown that microbial dysbiosis can affect local immune balance and cause inflammation [36, 37].

Fungi play an important role in maintaining the stability of bacteria and the entire microbiome. Studies have shown that adding specific fungi can promote the growth of certain bacteria [33], while fungi also play an important role in improving the adaptability of bacteria [38]. To further understand how abnormal fungal community status in CSD affects bacteria, leading to the disruption of local stability, we used LC/MS technique for metabolite detection. The metabolite analysis results showed differences in the metabolite spectra between the CSD and CON groups, and the regulatory mode of several different fungal species on metabolites was mainly negative regulation. By integrating fungal, bacterial, and metabolite information, we found that *Lachnellula suecica* played an important role in regulating the abundance of several *Lactobacillus* species. *Lachnellula suecica* reduces the abundance of several *Lactobacillus* species, including *Lactobacillus jensenii*, by decreasing the metabolites Goyaglycoside A and Janthitrem E. Meanwhile, *Clavispora lusitaniae* and *Ophiocordyceps australis* can also synergistically affect the abundance of *Lactobacillus* by influencing metabolite abundance.

So far, we have conducted a comprehensive analysis of bacterial and fungal communities and changes in environmental metabolites in Caesarean section scar diverticulum (CSD). Our results indicate that the local abnormal fungi in CSD affect the abundance of bacteria through specific metabolites, thereby destroying the stability of bacteria and the entire microbial community. This is an important supplement to the mechanism of the impact of CSD's local persistent inflammatory exudate on embryo implantation. It is difficult to achieve long-term efficacy with treatment plans targeting bacteria in CSD's treatment, which is a huge challenge in clinical treatment. The results of this study revealed that one of the important driving factors behind the local bacterial disorder in CSD is the abnormal composition and activity of fungi. This result has important implications for the update of clinical treatment plans. However, this study still has some limitations. This

study's lack of analysis of the antibiotic sensitivity of key fungi and bacteria can only provide preliminary ideas for clinical treatment. On the other hand, this study did not set up a negative control, so the discrimination effect on rare species is limited.

Declarations

Ethics approval and consent to participate

All study procedures were reviewed and approved by the ethics review board of the Sixth Affiliated Hospital of Sun Yat-Sen University (IRB no. 2019ZSLYEC-005S).

Authors' contributions

Xing Yang and Peigen Chen coordinated the study, participated in the design. Xing Yang carried out the study. Peigen Chen and Haicheng Chen analyzed and interpreted the data and drafted the manuscript. Xinyi Pan, Qianru Liu and Ziyu Liu recruit subjects. Xinyi Pan and Ziyu Liu participated in the follow-up work and data collection. Qianru Liu were also responsible for preparing ethical review materials. Xing Yang did the review of the final paper. All authors read and approved the final manuscript.

Competing interests

The authors declare that there are no conflicts of interest to disclose.

Data Availability

The metagenome sequencing has been deposited in China National Center for Bioinformation (<https://ngdc.cncb.ac.cn/>) under reference number PRJCA016850.

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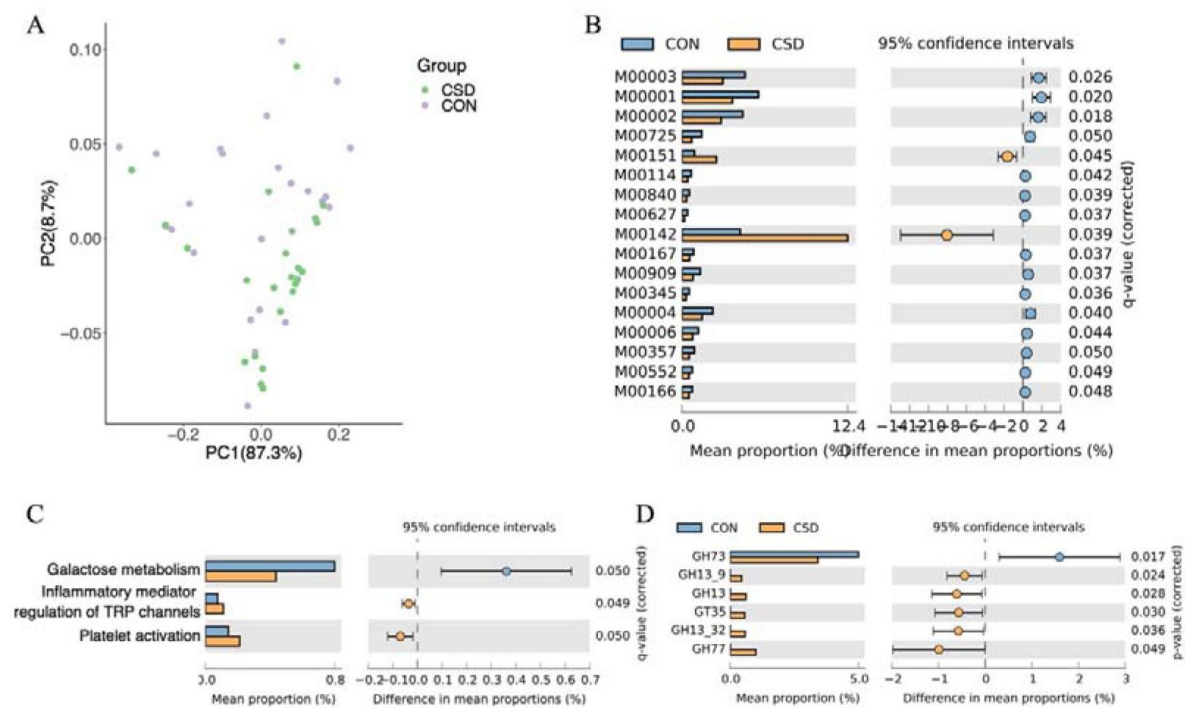


Fig S1.

Functional composition of the cervical microbiota. (A) PCA plot of microbial functional genes; (B) Differences in the composition of KEGG modules between CSD and CON; (C) Differences in the composition of KEGG pathway between CSD and CON; (D) Differences in the composition of carbohydrate-active enzymes between CSD and CON.

Correlation heatmap

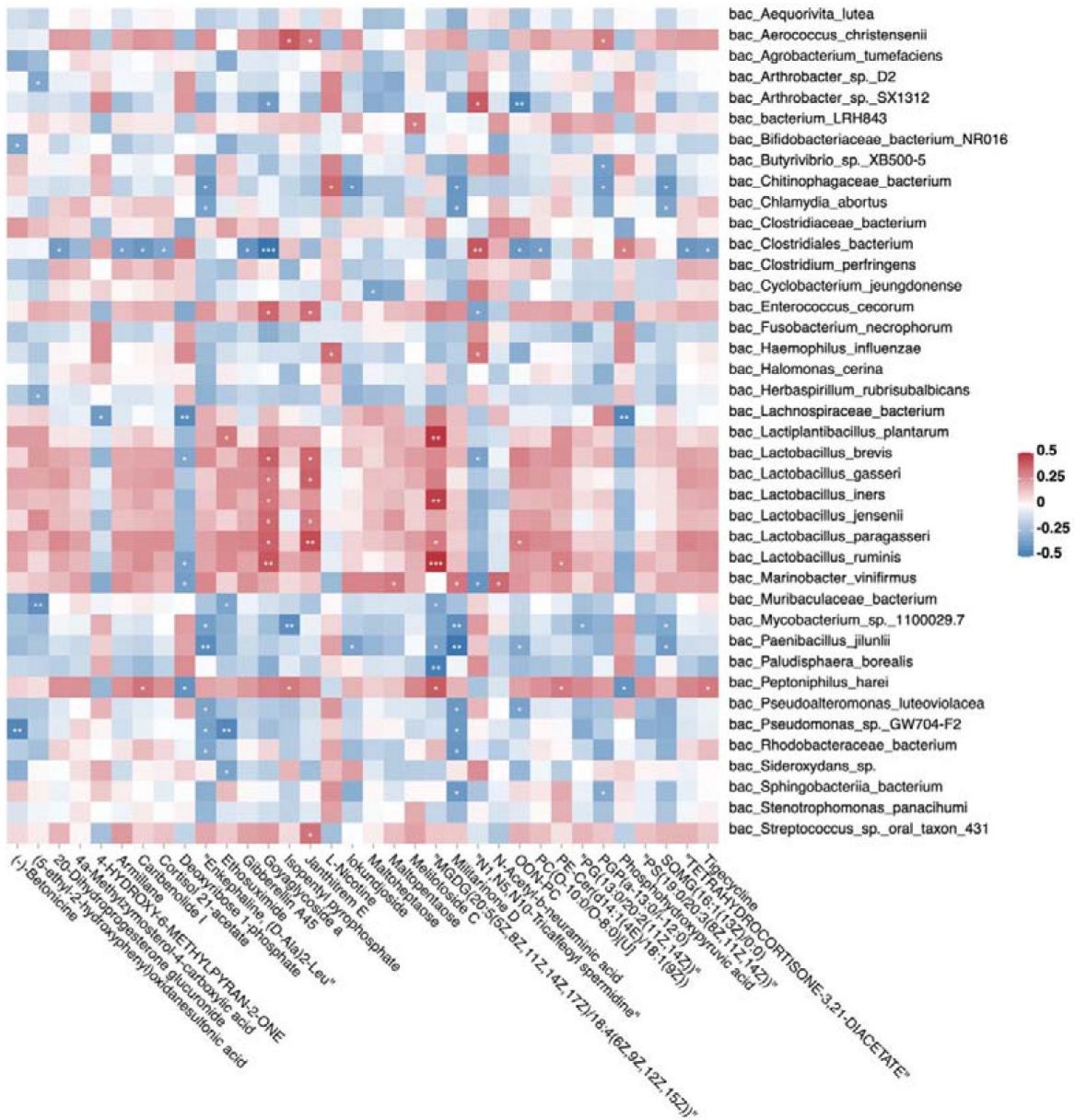


Fig S2.

Heatmap of the correlation between bacterial and metabolites.

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Editors

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University of California, Los Angeles, Torrance, United States of America

Reviewer #1 (Public Review):

Summary:

Chen et al. describe the bacterial and fungal composition of cervical samples from women with/without Cesarean-section scar diverticulum (CSD) using whole metagenomic sequencing. Also, they report the metabolomic profile associated with CSD and built correlation networks at the taxonomical and taxonomic-metabolic levels to establish potential bacteria-fungi interactions. These interactions could be used, long-term, as therapeutic options to treat or prevent CSD.

After reviewing the manuscript, the authors have not integrated any of my previous recommendations into the new version of the work. Therefore, in my opinion, the limitations or weaknesses of the study remain the same.

I find it especially worrying that they do not consider the use of white controls necessary, arguing that "we considered that this study described a biomass-rich site, and the abundance of dominant species was much higher than that of the possible 'kitome', so we did not set a blank control" while describing among the most predominant species in the reproductive tract bacteria that do not colonize humans and that have been previously described as contaminants.

Lack of experimental controls can lead to artifactual results and compromise the evidence presented and the significance of the results.

<https://doi.org/10.7554/eLife.90363.2.sa1>

Reviewer #2 (Public Review):

Summary:

Shotgun data have been analysed to obtain fungal and bacterial organisms abundance. Through their metabolic functions and through co-occurrence networks, a functional relationship between the two types of organisms can be inferred. By means of metabolomics, function-related metabolites are studied in order to deepen the fungus-bacteria synergy.

Strengths:

Data obtained in bacteria correlate with data from other authors.
The study of metabolic "interactions" between fungi and bacteria is quite new.
The inclusion of metabolomics data to support the results is a great contribution.

Weaknesses:

Most of them have been solved in the revision, but for the future it will be nice to integrate this data with others from 16s.

<https://doi.org/10.7554/eLife.90363.2.sa0>

Author Response

The following is the authors' response to the original reviews.

necessary clarifications on some of the reviewers' suggestions.

Reviewer #1 (Public Review):

Weaknesses:

- *This is a pilot study with only 24 cases and 24 controls. Because the human microbiota entails individual variability, this work should be confirmed with a higher sample size to achieve enough statistical power.*

Thank you for your suggestion. Unlike the high sparsity of 16s rRNA, the data density of metagenomic data is higher. Based on the experience of previous research, the sample size used this time can basically meet the requirements. However, your suggestion is very valuable, increasing the sample size allows better in-depth analysis. Due to limitations of objective factors, it is difficult for us to continue to increase the sample size in this study.

- *The authors do not report here the use of blank controls. The use of this type of control is important to "subtract" the potential background from plasticware, buffer or reagents from the real signal. Lack of controls may lead to microbiome artefacts in the results. This can be seen in the results presented where the authors report some bacterial contaminants (*Agrobacterium tumefaciensis*, *Aequorivita lutea*, *Chitinophagaceae*, *Marinobacter vinifirmus*, etc) as part of the most common bacteria found in cervical samples.*

Thank you for your suggestion. Applying blank controls in low biomass areas can effectively avoid contamination caused by the environment or kits. This opinion is consistent with that published by Raphael Eisenhofer et al. in Trends in Microbiology. When designing this study, we considered that this study described a biomass-rich site, and the abundance of dominant species was much higher than that of the possible 'kitome', so we did not set a blank control. On the other hand, our main discussion object in this study is high-abundance species, and the species filtering threshold for some analyzes was raised to 50%. Therefore, we believe that the absence of the blank control has little effect on the conclusions of this study. However, your opinion is spot on. Failure to set up a negative control will affect our future research on rare species. We will add a description in the Limitations section of the Discussion section.

- *Samples used for this study were collected from the cervix. Why not collect samples from the uterine cavity and isthmocoele fluid (for cases)? In their previous paper using samples from the same research protocol ((IRB no. 2019ZSLYEC-0055) they used endometrial tissue from the patients, so access to the uterine cavity was guaranteed.*

Thank you for your suggestion. In Author response image 1 we show the approximate location of our cervical swab sampling. There are two main reasons for choosing cervical swabs:

1. The adsorption of swabs allows us to obtain sufficient nucleic acid for high-depth sequencing, while the isthmocoele fluid varies greatly among patients, which will introduce unnecessary batch effects.
2. Since the female reproductive tract is a continuous whole, our sampling location is close to the lesion in the cervix, which can be effectively studied. On the other hand, the microbial biomass of the endometrium is probably two orders of magnitude lower than that of the cervix, and it is difficult to avoid contamination of the lower genital tract when sampling.

Based on the above reasons, we selected cervical swabs for our microbial data.

Author response image 1.



- *Through the use of shotgun genomics, results from all the genomes of the organisms present in the sample are obtained. However, the authors have only used the metagenomic data to infer the taxonomical annotation of fungi and bacteria.*

Thank you for your suggestion. The advantage of metagenomics is that it can obtain all the nucleic acid information of the entire environment. However, in the study of the female reproductive tract, the database of viruses and archaea is still immature, in order to ensure the accuracy of the results, we did not conduct the study. Looking forward to the emergence of a mature database in the future.

Reviewer #1 (Recommendations For The Authors):

- *It would be interesting to use another series of functional data coming from the metagenomic analyses (not only taxonomic) to expand and reinforce the results presented.*

Thank you for your suggestion. We have dissected the functional data of microbiota in the article.

- *The authors have previously published the 16S rRNA sequencing and transcriptomic analysis of the same set of patients. It would be nice to see the integration of all the datasets produced.*

Thank you for your suggestion. There is no doubt that integrating all the data will have more dimensional results. In our previous study we focused on microbe-host interactions. However, there is an unanswered question: What are the characteristics of the regulatory network within microbiota? Therefore, we answered this question in this study, exploring the complex interaction processes within microbial communities. In addition to direct effects, interactions between microbiota may also occur through special metabolite experiments. Therefore, we introduced the analysis of the untargeted metabolome. However, 16s rRNA can only provide bacterial information, so we did not integrate the data. In addition, the transcriptome provides host information and is not the focus of this study. However, your suggestion is very valuable, and we will integrate all the data in the next study on the exploration of treatment methods.

Reviewer #2 (Public Review):

Weaknesses: Methodological descriptions are minimal.

Some example:

**The CON group (line 147) has not been defined. I supposed it is the control group.*

- *There are no statistics related to shotgun sequencing. How many reads have been sequenced? How many have been removed from the host? How many are left to study bacteria and fungi? Are these reads proportional among the 48 samples? If not, what method has been used to normalise the data?*
- *ggClusterNet has numerous algorithms to better display the modules of the microbiome network. Which one has been used?*

Thank you for your suggestion. We have added details to the method.

Reviewer #2 (Recommendations For The Authors):

I think the author should take into account the points described in the "Weaknesses" section. The lack of detail extends to almost all the analyses that have been included in the manuscript. Although the results are sound, I think it is important to understand what has been analysed and how it has been analysed. It is important that all work is reproducible and this requires vital information.

For example, what parameters have been used for bowtie2? has a local analysis been used? or end-to-end ? Some parameters like --very-sensitive are important for this kind of analysis. You can also use specific programs like kneaddata.

The Raw data preprocessing section should be more detailed.

The same with the "Taxa and functional annotation" section, how have the data been normalised? has any Zero-Inflated Gamma probabilistic model algorithm been taken into account? How were the 0 (no species detected) in the shallow samples treated?

Which algorithms have been used for LEfSe ? Kluskal-Wallis->(Wilcoxon)->LDA ?

Which p-value has been used as cut-off ? this p-value has been corrected for multiple testing?

- *Information on ggClusterNet should be included and explained.*

The first section of the results and Table 1 should be in the Materials and Methods.

Thank you for your suggestion. We have added details to the method.

In the fungi section, it is mentioned that 431 species have been found. They should be included in a supplementary table.

How many bacteria were found? Please include them also in a supplementary table.

Thank you for your suggestion. We have added the corresponding table.

Reviewer #3 (Public Review):

Major

1. *Smoke or drink conditions, as well as diseases like hypertension and diabetes are important factors that could influence the metabolism of the host, thus the authors should add them in the exclusion criteria in the Methods.*

Thanks to reviewer #3 for professional comments. We have made corresponding additions in the method section. We also followed this standard when recruiting subjects.

1. *The sample size of this study is not large enough to draw a convincing conclusion.*

Thank you for your suggestion. Unlike the high sparsity of 16s rRNA, the data density of metagenomic data is higher. Based on the experience of previous research, the sample size used this time can basically meet the requirements. However, your suggestion is very valuable, increasing the sample size allows better in-depth analysis. Due to limitations of objective factors, it is difficult for us to continue to increase the sample size in this study.

Reviewer #3 (Recommendations For The Authors):

Please recruit more samples.

In addition, there are many formatting and grammatical mistakes in the manuscript.

Minor

1. *In Line 24-25 of the "Composition and characteristics of fungal communities", the format of "Goyaglycoside A and Janthitrem E." shouldn't be italic.*
1. *In Line 126 of the "Metabolite detection using liquid chromatography (LC) and mass spectrometry (MS)", the "10 ul" should be changed to "Ten ul". Beginning with arabic numerals in a sentence should be avoided.*

1. In Line 170 of the "Composition and characteristics of bacterial communities", the "162 differential species" should be "One hundred and sixty-two differential species".

1. In Line 187 of the "Composition and characteristics of fungal communities", the "42 differential" should be "Forty-two differential".

Thanks to reviewer #3 for professional comments. We have completely revised the language of the article.